Paolo Cremonese, Ph.D.

Vancouver, BC → Europe (Jul/Aug 2026 start) · Italian citizen — full EU work authorisation
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SUMMARY

Ph.D. physicist with a decade of research experience, specialising in **Bayesian statistical modelling, MCMC, uncertainty quantification, and large-scale data analysis** within the LIGO Scientific Collaboration. Track record of leading research end-to-end: problem formulation, model design, implementation, validation, and publication in top peer-reviewed journals. Hands-on experience with deep learning (TensorFlow, PyTorch), distributed HPC computing (Slurm, HTCondor), and production pipeline engineering. 35+ peer-reviewed publications, 1,300+ citations. Currently transitioning from academia to industry; target roles in quantitative research, data science, fintech, and AI/ML.

WORK EXPERIENCE

Physics Subject Matter Expert (Contract, part-time) Jan 2026 – present
Mercor *Remote*

- Evaluated AI-generated physics content for technical accuracy across a broad range of topics.
- Identified systematic failure modes in model outputs; provided domain-specific corrections.
- Collaborated with cross-functional teams to define and refine quantitative evaluation criteria and quality standards.

Postdoctoral Researcher — Lead Researcher, LIGO Scientific Collaboration Oct 2022 – Oct 2024
Universitat de les Illes Balears (UIB) *Palma de Mallorca, Spain*

- Led independent research programme on gravitational-wave lensing: detection, characterisation, and Bayesian parameter estimation of lensed events.
- **Built, tested, and validated Bayesian statistical models** for GW parameter estimation — assessing conceptual soundness, numerical implementation, and consistency with theoretical predictions; identified and corrected systematic biases.
- **Applied deep learning models** (TensorFlow, PyTorch) for signal detection across multi-TB astrophysical datasets, improving sensitivity benchmarks.
- **Co-led an automation project** for end-to-end lensing event detection, reducing manual analysis time by ∼70%.
- **Engineered and deployed runtomation**, a production pipeline automating Bayesian parameter estimation at scale — 100+ concurrent jobs on distributed HPC infrastructure (Slurm/HTCondor).
- Contributed to large international collaboration (1,000+ members); co-authored LIGO collaboration papers; presented at 10+ international conferences.

Ph.D. Researcher — Gravitational Wave Cosmology Oct 2018 – Oct 2022
University of Szczecin, Cosmology Group *Szczecin, Poland*

- Conducted original research on gravitational lensing degeneracies and their impact on gravitational-wave parameter estimation.
- Developed Bayesian and analytical models to tackle the Hubble tension problem using gravitational-wave standard sirens.
- Published 3 lead-author papers; presented at 5+ international conferences; co-organised one international conference.

Data Scientist Intern 2018
DBA Group *Treviso, Italy*

- Developed ML models for multi-domain datasets (meteorological, industrial) using scikit-learn and Python.
- Shipped production-grade predictive analytics tools to support data-driven business decisions.

EDUCATION

Ph.D. in Physics — Gravitational Lensing of Gravitational Waves Oct 2018 – Oct 2022
University of Szczecin *Poland*

Supervisor: Prof. Vincenzo Salzano. Cosmology Group (cosmo.usz.edu.pl). Thesis: lensing degeneracies, mass-sheet degeneracy breaking with GW interference, and Hubble constant constraints from lensed GW events.

M.Sc. in Astronomy Sep 2015 – Sep 2017
University of Padova / Stockholm University *Italy & Sweden*

Erasmus year at Stockholm University (2016–2017). Master thesis on gravitational lensing of gravitational waves (Stockholm).

B.Sc. in Astronomy Sep 2011 – Sep 2015
University of Padova *Italy*

Thesis on dark matter in spiral galaxies.

PUBLICATIONS

39 publications, 1,300+ citations (LIGO Collaboration member). Selected lead-author and key collaboration papers:

- **LensingFlow: An Automated Workflow for Gravitational Wave Lensing Analyses** (2025) — pipeline paper for the O4 lensing search.
- **Invariance transformations in wave-optics lensing: implications for GW astrophysics and cosmology** — Phys. Rev. D (2024).
- **Follow-up Analyses to the O3 LIGO-Virgo-KAGRA Lensing Searches** — MNRAS 526 (2023). LIGO Collaboration.
- **Characteristic features of GW lensing as probe of lens mass model** — Annalen der Physik (2021). 22 citations.
- **Breaking the mass-sheet degeneracy with gravitational wave interference in lensed events** — Phys. Rev. D 104 (2021). 63 citations.
- **High accuracy on H_0 constraints from gravitational wave lensing events** — Phys. Dark Universe (2019). 11 citations.

Full list: inspirehep.net/authors/1859874 · Google Scholar

Science communication: "Gravitational Waves and Deep Learning", Astrobites (Jul 2022). Divulgative talk: Dark energy and dark matter, YouTube (Oct 2022).

CONFERENCES & TALKS (15+)

- **"Mass-Sheet Degeneracy in Gravitational Wave Lensing"** — NBI Strong Lensing Group seminar, Copenhagen (Dec 2023); LVK Collaboration Meeting (Sep 2023).
- **"Wave Optics in Gravitational Wave Lensing"** — UBC Gravity Group seminar, Vancouver (Sep 2023).
- **"Breaking the mass-sheet degeneracy / GW lensing searches"** — GWverse (Sep 2021); Gravitex 2021 (Aug 2021); COSMO'21 (Aug 2021); Amaldi 14 (Jul 2021); Ibericos (Apr 2021).
- **Member of organising committee** — The 6th Conference of the Polish Society on Relativity, Szczecin (Sep 2019).

Full list: paolocremonese.com/CV#conferences

PROJECTS

runtomation *Python, Google Sheets API, Slurm/HTCondor*
Production pipeline automating the generation of setup files for cluster-scale Bayesian parameter estimation of GW signals. Deployed across 100+ concurrent HPC jobs; adopted by UIB lensing group.

paperbot *Python, ArXiv API, Telegram API*
Python package for systematic literature discovery: monitors ArXiv by keyword/author and delivers daily digests via Telegram. Actively used for research tracking.

Travel Buddy *Python, LLMs, RAG, full-stack, HTML/CSS*
Built during the VAM AI programme. Full-stack agentic AI application that generates personalised city-trip itineraries using LLM APIs, RAG pipelines, prompt engineering, and Google Maps integration.

myFinances *Swift, SwiftUI, C++, SQL, REST API*
iOS personal finance tracker: expense/income/investment logging, REST API, SQL backend (C++/Swift), SwiftUI frontend.

Spotify Playlist Manager *Python, Spotify API*
Automated tool for managing Spotify playlists and podcast queues via the Spotify API.

NoProfit Database *Python, AI web scraping, SQLite*
AI-powered scraper that builds and maintains a structured database of nonprofit organisations — built for use in Data For Good partnership work.

lensing *Python, Jupyter, numerical methods*
Research code for computing gravitational lensing of GW signals (developed with V. Salzano); public research artifact.

paolocremonese.com *HTML, CSS, JavaScript*
Personal website — hand-coded and self-maintained. Hosts CV, project pages, and blog.

SKILLS

Programming: Python (pandas, numpy, scipy, matplotlib, TensorFlow, PyTorch, scikit-learn, flask), SQL, C++, Swift/SwiftUI, HTML/CSS, JavaScript, Mathematica

ML & Statistics: Bayesian inference, MCMC, uncertainty quantification, stochastic modelling, deep learning, LLMs, RAG, prompt engineering, agentic AI

Infrastructure & Tools: Git/GitHub/GitLab, Linux, LaTeX, Slurm, HTCondor, Google Sheets API, REST APIs, Telegram

API, Spotify API

Languages: Italian (native), English (fluent), Spanish (basic), Polish (basic)

VOLUNTEERING

Partnerships Lead 2025
Data For Good *Vancouver, Canada*

Engaged nonprofit organisations to identify data needs and develop impactful project strategies. Built and deployed a CRM system to streamline client outreach and relationship management.

INTERESTS

Quantitative finance & investments · Deep learning & LLMs · Football & sports analytics · Running (half-marathons) · Piano · Philosophy & cosmology · Travel